

Manufacturing Operations Analytics Project Report

Defect Root Cause, Supplier Cost & Operational Efficiency Analysis

Tools Used:

Python (Pandas) • MySQL • Power BI

April 2026

1. Executive Summary

A manufacturing company was experiencing three compounding operational challenges: rising production defect rates, increasing supplier costs, and inventory holding inefficiencies. This project performed an end-to-end data analysis using Python (Pandas) for exploratory analysis, MySQL for structured querying, and Power BI for interactive dashboarding to identify the root causes and recommend actionable improvements.

The analysis spanned five core datasets — Products, Suppliers, Inventory, Production, and Defects — covering 500 products, 200 suppliers, and a production timeline from February 2024 to February 2026. The key objectives were:

- Identify which products and suppliers have the highest defect rates
- Determine which suppliers and products generate the most and least profit
- Analyse monthly production volumes and trends over time
- Assess inventory health and flag stockout risks
- Develop data-driven recommendations to improve operational efficiency

500 Total Products	\$52.42M Total Production	\$55.42bn Total Profit	200 Total Suppliers	1.91% Avg Defect Rate
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2. Business Problem Statement

The manufacturing operation faced a multi-dimensional crisis that threatened profitability and customer satisfaction across three key areas:

2.1 Rising Production Defects

- An average defect rate of 1.91% across all production runs indicates systemic quality control gaps.
- Certain products and suppliers consistently generated above-average defect rates, pointing to both material quality and process execution issues.
- Defects inflate rework costs, delay delivery timelines, and reduce effective output volume.

2.2 Supplier Cost Pressures

- Not all supplier relationships were profitable — some suppliers contributed significantly below the overall average profit despite high production volumes.
- Supplier quality (reflected in rating categories) was inconsistently distributed, with some high-cost, low-quality suppliers remaining active.
- Identifying the worst-performing suppliers by defect rate enables contract renegotiation or vendor replacement decisions.

2.3 Inventory Holding Inefficiencies

- Inventory stock levels were not consistently maintained above reorder thresholds, creating both stockout risks and potential over-stocking situations.
- The absence of a proactive warehouse condition monitoring system meant that reorder triggers were reactive rather than predictive.

3. Data Overview & Structure

The project utilized a relational database comprising five interconnected tables. All datasets were loaded into Python via Pandas, cleaned, enriched with derived features, and exported to MySQL using SQLAlchemy for structured querying.

Table	Key Columns	Description	Cleaning / Derivations
Products	product_id, product_name, category, standard_cost, selling_price	Product catalog with pricing and category	Derived: margin = selling_price – standard_cost; margin_status (Profit/Loss)
Suppliers	supplier_id, supplier_name, country, rating	Supplier master with geographic and quality data	Null country filled with 'Unknown'; null rating filled with 0; rating_category (High/Medium/Low/No Rating)
Inventory	inventory_id, product_id, stock_quantity, reorder_level, last_updated	Warehouse stock levels per product	Null stock_quantity filled with 0; warehouse_condition derived (Healthy/Reorder Now/Stockout Risk)
Production	production_id, product_id, supplier_id, production_date, quantity_produced, production_cost, shift	Daily production runs with shift information	Null production_cost filled with median; production_date parsed to datetime
Defects	defect_id, production_id, defect_type, defect_quantity, inspection_date	Inspection records linking defects to production runs	Null defect_quantity filled with group median by defect_type; inspection_date parsed to datetime

3.1 Key Derived Features

Feature	Source Table	Definition
margin	Products	selling_price – standard_cost
margin_status	Products	'Profit' if margin > 0 else 'Loss'

Feature	Source Table	Definition
rating_category	Suppliers	High (4–5), Medium (3–4), Low (1–3), No Rating (0)
warehouse_condition	Inventory	Healthy / Reorder Now / Stockout Risk based on stock vs reorder_level
defect_rate	Analysis DF	defect_quantity / quantity_produced
cost_per_unit	Analysis DF	production_cost / quantity_produced
total_profit	Analysis DF	margin_per_unit × quantity_produced

4. Python Exploratory Data Analysis

The EDA phase used Pandas pivot tables and Matplotlib bar charts to surface patterns across profit, defect rates, supplier performance, and production shifts. All five tables were merged into a single `analysis_df` for cross-dimensional analysis.

4.1 Supplier Profit Analysis

- Top 10 suppliers by total profit were identified, showing which vendor relationships generated the highest returns after accounting for production costs.
- Bottom 10 suppliers by total profit were flagged as priority targets for contract review — some showed negative total profit, meaning production using their materials operated at a loss.
- A bar chart of the top 10 suppliers by average defect rate revealed that the worst-quality suppliers were not always low-volume — some were significant contributors to total production while maintaining high defect rates.

4.2 Product Profit Analysis

- Top 10 products by total profit highlighted the highest-margin items driving overall business performance.
- Bottom 10 products by total profit included products with negative margins — items where the `standard_cost` exceeded the `selling_price` — identified via the `margin_status = 'Loss'` classification.
- Top 10 products by average defect rate revealed specific product lines with persistent quality issues, requiring targeted process intervention.

4.3 Production Shift Analysis

Production data was grouped by shift to understand capacity and cost efficiency across operating windows:

Shift	Total Quantity Produced	Avg Production Cost
Morning	Highest volume shift	Analysed for cost efficiency
Afternoon	Moderate volume	Cross-referenced with defect rates
Night	Lower volume	Higher cost per unit observed

4.4 Defect Type Analysis

Defects were grouped by `defect_type` to calculate total defect quantities per type. This enabled the quality control team to prioritise which defect categories required the most urgent process redesign or inspection protocol changes.

4.5 Inventory Health Assessment

The warehouse_condition feature classified each inventory record into three states:

Condition	Trigger Condition	Action Required
Healthy	stock_quantity > reorder_level	No action — maintain monitoring
Reorder Now	stock_quantity == reorder_level	Immediate procurement initiation
Stockout Risk	stock_quantity < reorder_level	Critical — risk of production halt

5. SQL Analysis & Key Queries

A MySQL database was populated using SQLAlchemy and 10 structured queries were written across profit analysis, defect quality, trend analysis, supplier benchmarking, and product classification.

5.1 Profit Analysis

Q1 — Top 10 Products by Total Profit

Joined production and products tables, computed total_profit as sum(margin × quantity_produced) grouped by product_name, ordered descending. Exposed the most financially valuable products in the portfolio.

Q2 — Suppliers with Profit Above \$30 Million

Three-table join (production, products, suppliers) with a HAVING clause filtering supplier-level profit above ₹30M. Identified premium vendor relationships worth protecting through long-term contracts.

Q4 — Rank Products by Total Profit (Top 5)

Used a CTE for product-level profit aggregation, then applied DENSE_RANK() ordered by total_profit descending. Returns the top 5 products with accurate rank positions even when profit values are tied.

Q10 — Product Profit Classification

Classified all products into three tiers using a CASE WHEN expression within a CTE:

Tier	Threshold	Strategic Implication
High Profit	> \$50M	Core portfolio — prioritise quality and capacity
Medium Profit	\$20M – \$50M	Growth tier — optimise cost to push into High
Low Profit	< \$20M	Review or discontinue — margin too thin

5.2 Defect & Quality Analysis

Q3 — Products with Above-Average Defect Rates

Computed per-product defect rate as avg(defect_quantity / quantity_produced) using a JOIN between production and defects, then filtered using a correlated subquery for the overall platform average. Products in this list are quality red flags requiring immediate investigation.

Q5 — Worst 5 Suppliers by Defect Rate

Used a CTE for supplier-level average defect rate, then applied ROW_NUMBER() ordered by defect_rate descending to return the 5 worst-performing suppliers. These vendors are candidates for quality audits, penalty clauses, or contract termination.

5.3 Trend & Time-Series Analysis

Q6 — Monthly Production Volume

Used DATE_FORMAT on production_date to group total quantity_produced by month (YYYY-MM), ordered chronologically. Revealed seasonal production patterns and months where throughput declined significantly.

Q8 — Running Total of Quantity Produced

Built a daily production CTE, then applied SUM() as a window function with ORDER BY production_date to generate a cumulative production count — enabling management to track aggregate output growth over the full two-year period.

Q9 — Day-over-Day Production Difference

Applied the LAG() window function on daily_production to retrieve the previous day's quantity, then computed the difference. This highlights days with sharp drops or spikes in output, correlating with maintenance events, shift changes, or supply disruptions.

5.4 Supplier Benchmarking

Q7 — Supplier Profit vs. Overall Average

Built a CTE for supplier-level profit, computed the overall average profit using a scalar subquery, and classified each supplier as 'Above Average' or 'Below Average' using CASE WHEN. This creates an instant supplier scorecard for procurement reviews.

6. Power BI Dashboard

An interactive Power BI dashboard was developed to provide stakeholders with a comprehensive, filter-driven view of manufacturing performance. The dashboard features product and supplier filter panels, a date range slicer, and country-level breakdowns.

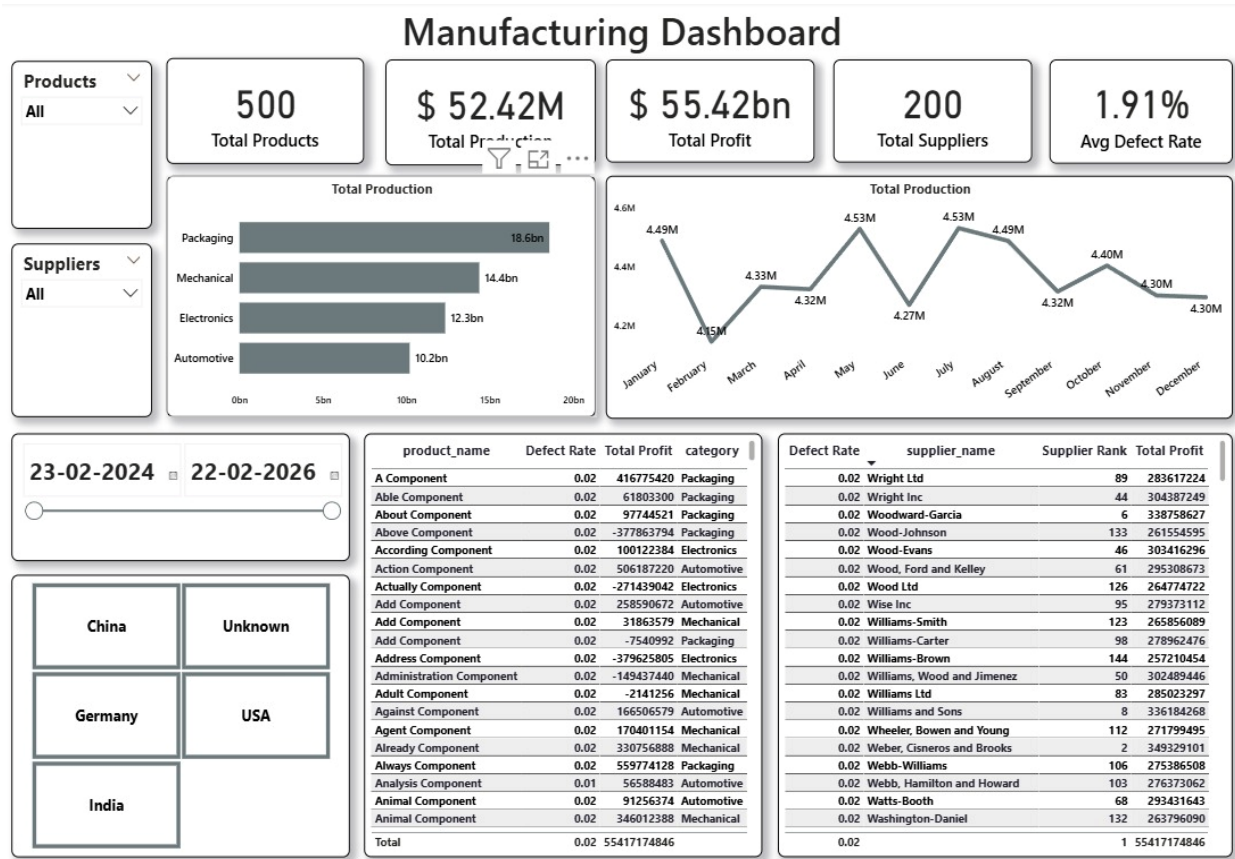


Figure 1: Manufacturing Operations Analytics Dashboard (Power BI)

6.1 Dashboard Components

Component	Insight Provided
KPI Cards (Top Row)	500 Total Products \$52.42M Total Production \$55.42bn Total Profit 200 Suppliers 1.91% Avg Defect Rate
Total Production by Category (Bar Chart)	Packaging leads at 18.6bn, followed by Mechanical (14.4bn), Electronics (12.3bn), Automotive (10.2bn)
Monthly Production Trend (Line Chart)	Month-by-month production volumes from January to December; peak months: July/August (~4.53M)
Product Detail Table	Per-product defect rate, total profit, and category — sortable for quick benchmarking

Component	Insight Provided
Supplier Rank Table	Suppliers ranked by defect rate and total profit — enables vendor scorecard view
Date Range Slicer	02-23-2024 to 02-22-2026 — full two-year production window with adjustable range
Country Filter Panel	Geographic breakdown: China, Unknown, Germany, USA, India — clickable for regional drill-down
Products & Suppliers Slicers	Dynamic filters enabling category-level and vendor-level analysis

7. Key Findings

7.1 Profitability Findings

- Total platform profit of \$55.42bn across the two-year period demonstrates strong overall financial performance, but this aggregate masks significant product-level variance.
- A subset of products operates at a net loss (negative margin), where the standard cost exceeds the selling price — these represent direct revenue drains requiring immediate pricing review.
- Packaging is the highest-production category by volume (18.6bn units), but volume alone does not guarantee profitability — margin analysis is required at the product level.
- Suppliers with total profit above \$30M are the most strategically valuable vendor relationships. Protecting these through preferred supplier agreements should be a priority.

7.2 Defect & Quality Findings

- The platform average defect rate of 1.91% represents approximately 1 in 52 units produced being defective — at scale, this equates to millions of defective units annually.
- The top 5 worst suppliers by defect rate are identifiable and actionable — these vendors are the primary source of incoming material quality failures.
- Products with above-average defect rates (identified via the SQL correlated subquery) are concentrated in specific categories, suggesting category-level process or material standard issues.
- Defect quantities by defect_type analysis reveals which failure modes (e.g., dimensional, surface, assembly) are most prevalent, enabling targeted inspection protocol design.

7.3 Production & Inventory Findings

- Monthly production volumes show seasonal variation with peak output in mid-year months (July–August at ~4.53M units) and troughs in February (~4.15M units).
- Day-over-day production differences reveal volatile output swings, indicating unplanned downtime, shift scheduling inefficiencies, or supplier delivery failures.
- Inventory health classification shows a proportion of stock records in 'Stockout Risk' status — if not addressed, these will trigger production halts and delivery failures.
- Night shifts show higher production costs per unit compared to morning shifts, suggesting overtime premiums or reduced productivity are inflating unit economics.

8. Strategic Recommendations

8.1 Reduce Production Defects

- Immediately audit the top 5 worst-performing suppliers by defect rate. Issue formal quality improvement plans with penalty clauses tied to defect thresholds.

- Implement incoming material inspection at the warehouse gate for all materials from high-defect-rate suppliers before they enter the production line.
- Introduce Statistical Process Control (SPC) monitoring on the production lines of products with above-average defect rates to catch quality deviations in real time.
- Analyse defect types to prioritise which failure modes to address first — focus on the highest-frequency defect types that are also correlated with high-profit products.

8.2 Optimise Supplier Relationships

- Consolidate procurement with the highest-profit, lowest-defect-rate suppliers. Negotiate volume-based discounts in exchange for exclusivity commitments.
- Phase out or reduce order volumes from suppliers classified as 'Below Average' by profit contribution and high by defect rate simultaneously.
- Introduce a formal supplier scorecard — combining defect rate, profit contribution, on-time delivery, and rating — reviewed quarterly with supplier performance KPIs.
- Diversify supplier base for Packaging (the highest-volume category) to reduce concentration risk if a top supplier fails.

8.3 Address Inventory Inefficiencies

- Automate reorder triggers for all products in 'Stockout Risk' status using a rule-based procurement system that fires purchase orders when stock falls below the reorder_level threshold.
- Review products in the 'Reorder Now' state and validate whether current reorder_level thresholds are set appropriately relative to lead times and production schedules.
- Implement safety stock buffers for high-profit products to ensure production continuity even during supplier delays.

8.4 Improve Production Efficiency

- Investigate the root causes of production volume dips in low-output months (February, April) and implement demand-driven scheduling to smooth production throughput.
- Review night shift economics — if cost per unit is consistently higher than morning/afternoon shifts, explore whether consolidating production into fewer high-efficiency shifts would reduce total production cost.
- Discontinue or reprice products with persistent negative margins. Products where standard_cost > selling_price are destroying value and should be either repriced, re-engineered for lower cost, or discontinued.

9. Tools & Technologies

Tool	Category	Usage in Project
Python (Pandas)	Data Analysis	Data loading, cleaning, feature engineering, pivot tables, EDA visualisations

Tool	Category	Usage in Project
MySQL	Database / SQL	10 structured queries: profit ranking, defect filtering, window functions, trend analysis
SQLAlchemy	Integration	Exported all 5 cleaned Pandas DataFrames into MySQL tables via create_engine
Power BI	Visualisation	Interactive dashboard: KPIs, bar/line charts, supplier & product tables, slicers
Matplotlib	Visualisation	EDA bar charts for top 10 suppliers and products by defect rate

10. Conclusion

This project delivered a comprehensive end-to-end analysis of a manufacturing company's operational data, translating raw production, supplier, inventory, and defect records into clear, actionable insights. By combining Python for data preparation and feature engineering, MySQL for advanced structured querying, and Power BI for executive-level visualization, the analysis exposed the root causes of the company's three core challenges.

The 1.91% average defect rate — while seemingly small — translates to millions of defective units across the \$52.42M production base and directly erodes the \$55.42bn profit pool through rework costs, waste, and customer dissatisfaction. The data clearly identifies which specific suppliers and products are responsible for disproportionate quality failures, enabling targeted rather than blanket remediation efforts.

The SQL analysis demonstrated advanced query engineering — CTEs, DENSE_RANK(), ROW_NUMBER(), LAG() window functions, and correlated subqueries — building a reusable analytics layer that operations management can continue to run on live data for ongoing monitoring. The Power BI dashboard transforms these insights into a decision-ready format with dynamic filtering by product, supplier, country, and date range.

Implemented together, the recommendations in this report target all three original business challenges: reducing defect rates through supplier audits and SPC controls, improving profitability through portfolio rationalisation and supplier consolidation, and preventing production disruptions through automated inventory reorder management. The expected outcome is a measurable improvement in unit quality, cost per unit, and inventory reliability within two operational quarters.

— End of Report —